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# Project Name

**Connecting the Last Mile — Internet for All Territorians**

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## CONCEPT

A hybrid solution addressing the Northern Territory's digital divide by:

1. Developing an **offline-first learning app** to give remote students access to educational resources without continuous internet.
2. Deploying **community-based mesh networks** to provide affordable, reliable internet coverage in underserved areas.

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## PERSONAS

- **Remote Students** – Children and young adults in rural and Indigenous communities without stable internet access.
- **Teachers & Educators** – Upload and distribute learning materials that students can use offline.
- **Healthcare Workers** – Deliver telehealth services with low data usage.
- **Community Leaders & Councils** – Oversee deployment and sustainability of mesh networks.
- **Local IT Volunteers** – Assist in setup, troubleshooting, and maintenance.

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## CONTEXT OF USAGE & COVERAGE

- **Offline-first** mode for learning app with local data storage.
- Syncs only when a 4G or Wi-Fi connection is available.
- Mesh network nodes extend coverage across the community using a **single satellite or cellular uplink**.
- Designed for **low-bandwidth, high-latency** environments common in NT's remote areas.

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## STAKEHOLDERS

- **NT Government – Digital Strategy & Education Departments** – Provide funding and policy support.
- **Community Councils** – Approve and manage deployments.
- **Schools & Training Centres** – Integrate app into curriculum.
- **Healthcare Services** – Use connectivity for telehealth.
- **Technology Partners** – Supply routers, uplink, and app development support.

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## COMPONENTS

1. **Offline Learning App** – Educational content, quizzes, video playback, and auto-sync.
2. **Mesh Network Hardware** – Low-cost routers, Raspberry Pi nodes, and satellite uplink.
3. **Management Dashboard** – For network monitoring, bandwidth allocation, and troubleshooting.

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## FEATURES

- Works fully offline with sync-on-connect capability.

- Video compression and adaptive streaming to save bandwidth.
- Automatic mesh routing for uninterrupted local connectivity.
- Shared bandwidth among households to reduce costs.
- Easy network setup with plug-and-play nodes.

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## MILESTONES

1. **Research & Planning** – Identify target community and design network architecture (2 weeks).
2. **Prototype Learning App** – Core offline content and sync function (4 weeks).
3. **Simulate Mesh Network** – Lab-based testing and optimisation (2 weeks).
4. **Pilot Deployment** – Install in one NT community, gather feedback (6 weeks).
5. **Refinement & Expansion Plan** – Incorporate feedback and prepare Code Fair presentation.

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## DELIVERABLES

- Progressive Web App (PWA) or APK for learning platform.
- Mesh network simulation and hardware setup documentation.
- Poster and presentation for CDU IT Code Fair 2025.
- Community training materials.

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## TECHNOLOGY

- **Frontend:** Flutter or React for PWA.

- **Backend:** Node.js with offline sync API.
- **Network Simulation:** NS-3 or Cisco Packet Tracer.
- **Hardware:** Raspberry Pi, Ubiquiti routers.

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## PLATFORM, OS

- Android smartphones/tablets (primary).
- Web browsers (secondary).

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## ORIENTATION

- **Both** portrait and landscape support.

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## RISKS

- Difficulty sourcing affordable, durable hardware.
- Limited local technical expertise for maintenance.
- Weather and environmental challenges for outdoor equipment.
- Funding sustainability for long-term operations.

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## PROBLEMS TO SOLVE

- Inconsistent or non-existent internet in remote NT.
- High data costs making internet unaffordable.

- Education and telehealth access gaps for remote residents.

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## RELEASE

- Pilot in selected NT communities via a **local council partnership**.
- Public release as a **free app** with open-source mesh network guidelines.

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## OBJECTIVES & PURPOSES

- Reduce the **digital access gap** in pilot communities by **at least 30%** within the first year.
- Improve student access to curriculum content and telehealth services.
- Create a scalable model for other rural/remote regions.

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## VALUES

1. **Accessibility** – Inclusive design for all users, regardless of connectivity.
2. **Affordability** – Shared infrastructure lowers costs for individuals.
3. **Community Empowerment** – Locally managed and maintained networks.

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